Eric Liu (劉佳衛)

Technical Project Consultant & Enterprise Solutions Architect

📍 Taipei, Taiwan ⇄ Ho Chi Minh City, Vietnam 📧 jiacyu@gmail.com 📱 +886-916-100-757 🌐 Decision Systems Portfolio ↗



EXECUTIVE SUMMARY

1. **Enterprise Systems Architecture & Integration:** Over 12 years bridging complex commercial operations with cutting-edge digital infrastructure. Proven track record architecting multi-module business systems, localized IoT automation networks, and air-gapped decision platforms.
2. **Zero-Cloud Client-Side Reactive Applications:** Pioneer in developing high-density, client-side single-page applications (SPAs) that execute complex parametric sensitivity engines purely in the browser—guaranteeing 100% data confidentiality for enterprise and private equity clients.
3. **Dual-GPU Orchestration & VRAM Arbiter:** Engineered cross-process Win32 named mutex locks, dynamic memory swapping, and automated leak-recovery watchdogs across dual-GPU hardware, enabling high-throughput parallel inference while eliminating GPU Out-Of-Memory crashes.
4. **Code-Driven Deterministic Media Pipelines:** Created zero-disk I/O video rendering pipelines integrating GSAP deterministic stepping, Puppeteer Headless direct streaming (Direct Pipe), and neural audio synthesis, automating visual research report production.
5. **Database Performance & Infrastructure Tuning:** Hands-on expertise in relational database schema design, query optimization (achieved 60% query latency reductions in manufacturing ERPs), Python microservices, and automated cron pipelines.

CORE COMPETENCIES & TECHNICAL CAPABILITIES

- **Enterprise Architecture & Web Systems**
  - Zero-Cloud SPA Architecture
  - 60FPS Reactive State Engines
  - Chart.js Data Visualization
  - FastMCP Microservices Protocol
  - RESTful API Integration
  - Cross-Platform Deployment
- **Dual-GPU Orchestration & AI Inference**
  - Dual-GPU Acceleration
  - Win32 Named Mutex Concurrency
  - Dynamic VRAM Page Swapping
  - VAEDecodeTiled Memory Conservation
  - On-Premise GPU Whisper
  - Offline Neural Translation (Hy-MT2)
- **Data Infrastructure & Performance**
  - SQL Query Optimization (-60%)
  - Relational Schema Engineering
  - Python Microservices
  - Linux / WSL2 Environments
  - Git Version Control & CI/CD
  - Automated Cron Pipelines
- **Commercial Governance & Delivery**
  - Cross-Border Entity Deployment
  - Statutory Compliance (Decree 09)
  - Async Technical Documentation
  - Vendor Contract Negotiation
  - Bilingual English / Mandarin
  - B.S. in Quantitative Finance

PROFESSIONAL EXPERIENCE (PART 1: SENIOR EXECUTIVE LEADERSHIP)

Free Link System / Wake N' Bake

2015 - 2020

Founder & Managing Director — Taipei, Taiwan

- Founded premier smart building IoT engineering firm, specializing in Modbus, Zigbee, and Z-Wave cross-brand protocol integration for high-end residential and commercial developments.
- Secured master distribution rights for Fibaro (Poland) and Wulian (China); distributed HP Enterprise/Aruba networking hardware to enterprise clients and retail chains.
- Formulated and implemented standardized engineering specifications for Tier-1 real estate developers, architectural firms, and certified electrical contractors.
- Designed and maintained proprietary ERP and inventory relational database schema, connecting hardware procurement, warranty management, and customer CRM modules.

Locus Automation JSC / Tech-Link Silicones Vietnam

2020 - 2022

General Manager & Group Systems Director — Ho Chi Minh City, Vietnam

- Directed technology solutions and commercial systems for a subsidiary of a 1,300-headcount manufacturing group with 9 factories across China and Vietnam.
- Supervised a 34-person engineering and business development team, restructuring production reporting pipelines to achieve net profitability in 6 months.
- Overhauled factory inventory and SQL query architecture, accelerating multi-plant daily ERP reporting from 45 minutes to 18 minutes (60% latency reduction).
- Delivered enterprise smart conference and automation systems to dozens of prominent Taiwanese industrial plants and top-tier Vietnamese corporations (Bitis, Minh Long).

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PROFESSIONAL EXPERIENCE (PART 2: COMMERCIAL DISTRIBUTION & BANKING)

CEREALOUTFIT CO., LTD
2019 - Present

Director of Technology & Operations
Taipei, Taiwan ⇄ Ho Chi Minh City, Vietnam

- Architected omni-channel e-commerce systems, database schemas, and digital marketing tracking infrastructure supporting NT$ 52M annual turnover and 164,000 social fans.
- Led Vietnam technical market entry, securing Ministry of National Defense electronic commerce security permits and deploying compliant local server and payment gateways.
- Engineered automated inventory replenishment and financial health analytics pipelines in Python and SQL, maintaining zero downtime across regional peak sales.
- Spearheaded technical vendor evaluations and cloud cost optimization initiatives, reducing monthly infrastructure expenditure while supporting 15,000 monthly active users.

Citi Bank NA & Bank SinoPac
2011 - 2015

Junior Manager — Wealth Management & SME Banking
Taipei, Taiwan

- Managed CitiGold wealth management portfolios and structured SME credit facilities, developing analytical cash flow models for corporate clients.
- Conducted rigorous financial risk evaluations and statutory compliance audits for cross-border commercial borrowers.
- Maintained spotless audit records and repeatedly ranked in top quarterly performance tiers, establishing deep grounding in financial engineering and corporate governance.

FLAGSHIP QUANTITATIVE & ENTERPRISE DECISION SYSTEMS

Zero-Cloud Institutional Decision Applications — Principal Systems Architect
`JavaScript, Tailwind, Chart.js, Reactive State Machine`
Architected suite of institutional decision applications that execute complex financial models purely client-side with zero external cloud dependencies or data exposure risks. (**Impact:** Delivered C-suite grade modeling tools with instantaneous reactive sliders and mathematical transparency.)

Dual-GPU Orchestration & VRAM Arbiter — Core Systems Engineer
`Python, Win32 Named Mutex, PyTorch, CUDA, Dual-GPU`
Designed cross-process named mutexes, dynamic VRAM page swapping, and leak-recovery watchdogs across dual-GPU hardware, enabling parallel multi-LLM execution and sub-second task handover. (**Impact:** 0% VRAM collision, eliminating GPU OOM crashes and slashing enterprise on-premise hardware infrastructure costs.)

Code-Driven Deterministic Media Pipelines (Web-to-Video) — Media Tech Architect
`GSAP, Puppeteer, Direct Pipe, FFmpeg, Python, Neural TTS`
Pioneered a code-driven video synthesis pipeline combining GSAP deterministic stepping, Puppeteer Direct Pipe zero-disk streaming, and neural voice synthesis, achieving frame-accurate alignment. (**Impact:** Accelerated visual research reporting efficiency by 90%, completely replacing manual post-production video editing.)

Vietnam HCMC Ocean Freight Intelligence Platform — Data Systems Consultant
`SCFI Trajectory Engine, Carrier Reliability Matrix, Dynamic Search`
Designed strategic intelligence platform evaluating Southern Vietnam's $138.4B export hinterland, carrier schedule reliability, and freight procurement hedging protocols. (**Impact:** Empowered corporate shippers to index long-term ocean contracts against dynamic spot-rate trajectories.)

EDUCATION

National Dong Hwa University
Bachelor of Science (B.S.) in Quantitative Finance & Banking
2005 - 2009 · Major in Quantitative Finance & Banking

PROFESSIONAL FINANCE & REGULATORY LICENSES

- **Financial Derivatives Specialist License** — Securities & Futures Institute (SFI)
- **Trust Operations & Planning Specialist License** — Taiwan Academy of Banking and Finance
- **Senior Life Insurance & Investment-Linked Products** — Life Insurance Association (LIA)
- **Property & Casualty Underwriting License** — Non-Life Insurance Association
- **Securities Senior Representative & Wealth Advisory** — Securities Association